STUDENT ID NUMBER _____

UNIVERSITY OF OTAGO EXAMINATIONS

COMPUTER SCIENCE

Paper COMP160

GENERAL PROGRAMMING

Sample Final Exam No. 1

(TIME ALLOWED: TWO HOURS)

This examination paper consists of 16 pages including this cover page. **Please check that you have a complete set.** The total and maximum mark for this paper is 80.

Candidates **may** bring a single A4 sheet of notes (both sides) into the exam as reference material (no magnifying devices are allowed). Candidates **may not** bring in any other written material, calculators or electronic devices.

This exam is divided into multiple choice and short answer sections.

For the multiple choice questions:

Fill in your name and Student ID number on the form provided for your answers. Fill in the correct ovals for your Student ID. Put all answers on the form and use your general answer book for any rough work. Each multiple choice question is worth one mark. Select only one answer per question which you think is the BEST answer.

When marking your answers, **make sure you fill in the oval completely.** Remember your answer form is machine marked. Failure to complete the answer sheet correctly may result in lost marks. If you wish to change your choice, erase the original answer completely.

For the short answer questions:

Fill in your Student ID number on the top of each page of short answer questions in this script. Write all your answers in the spaces provided in this script, and use your general answer book for any rough work. Short answer questions are worth various marks as indicated, for example: (3). For any of these questions part marks may be given.

In this paper we have tried to represent some of the DrJava syntax colouring using **bold**, but this is for information only and mistakes in the use of bold or fonts should not be taken to be mistakes in the programs or script.

Please write your ID number in the top right hand corner of this page and the back page.

When you are finished hand in the answer sheet, this script, and your general answer book. All must have your ID number filled in.

TURN OVER

Multi-choice section. Please answer on the sheet provided.

1. What is the Java platform?

   (A) The price of coffee beans as traded on the international exchange.
   (B) The standards and tools which support the Java language.
   (C) The policies and manifesto of the Legalise Java Party.
   (D) An automatic process which frees up unused memory.
   (E) The bench top that an espresso machine sits on.

2. Which of the following is **not** a feature of good programming practice?

   (A) Plan your code before you start typing.
   (B) Use meaningful names for variables and methods, and follow Java naming conventions.
   (C) Add comments to your code when somebody else wants to read it.
   (D) Test your code carefully.
   (E) Use clear code that others might understand readily.

3. Reasons for having different data types include:

   (A) Allowing characters to be ordered before numbers.
   (B) Allowing the interpreter to run on different kinds of hardware platform.
   (C) Allowing for programming in different languages.
   (D) Allowing the programmer to exactly control the format of the program output.
   (E) Allowing the compiler to check for consistency and use memory efficiently.

4. Which of the following are two keywords in Java which are often used to save duplication by having constructors call other constructors?

   (A) public, void
   (B) static, this
   (C) abstract, final
   (D) this, super
   (E) extends, super

5. Which of the following statements is **false**?

   (A) Two or more methods declared in a class can have the same name.
   (B) Two or more methods declared in a class can have the same signature.
   (C) We can specify the visibility of data fields and methods.
   (D) Two or more methods declared in a class can have the same parameter list.
   (E) The order that methods are declared in a class is not important.

6. Which of the following is the best definition of encapsulation (an OO design concept)?

(A) All inputs required by a method should be passed as parameters.
(B) Names for data fields and methods should reflect their functions.
(C) Data should be directly accessed only by the object that contains it.
(D) Packages should contain classes that are related to each other.
(E) The degree to which different classes call each other's methods should be minimised.

7 What is achieved by the statement below?

double [ ] dip;

(A) It declares a variable `dip` of type array of double.
(B) It declares and initializes a variable `dip` of type array of double.
(C) It declares a variable `dip` of type double.
(D) It declares and initializes a variable `dip` of type double.
(E) It creates an instance of the class double.

8. Which of the following is **false**?

(A) The loop control variable of a for loop must be incremented by 1.
(B) A for loop header specifies an initialisation, a condition, and an update.
(C) A do..while loop always executes the loop body at least once.
(D) A for loop can always be re-written as a while loop.
(E) A while loop might never execute the loop body.

9. A for loop is supposed to process every element in an array called data. Which of the following is the most appropriate header for the loop?

(A) **for** (**int** i = 0; i < data.length; i++)
(B) **for** (**int** i = 0; i <= data.length; i++)
(C) **for** (**int** i = 0; i == data.length; i++)
(D) **for** (**int** i = 1; i < data.length; i++)
(E) **for** (**int** i = 1; i <= data.length; i++)

10. The code below declares an enumerated type. How is it implemented in Java?

```
enum Flower {rose, tulip, daffodil, daisy, buttercup}
```

(A) As an array with elements that refer to objects of type String.
(B) As an array with elements that refer to objects of type enum.
(C) As a class with methods that return references to objects of type enum.
(D) As a class with static data fields that refer to instances of type Flower.
(E) As an interface with static data fields that refer to the class object Flower.

11. Which of the following statements about visibility is **false**?

(A) Private members are visible only within the class where they are defined.
(B) If no visibility is specified the default is package visibility.
(C) We can specify the visibility of classes and members of classes.
(D) If a class is public all its members are automatically public too.
(E) Protected members are visible to classes extending the class in which they are defined.

12. A program declares a data field as follows: `AClass a = new AClass( );`
What kind of value does the data field a hold?

(A) A class of type AClass.
(B) An object of type AClass.
(C) A reference to an object of type AClass.
(D) A reference to an array of type AClass.
(E) A reference to an array of references to objects of type AClass.

13. Which of the following is an important property of class / static data fields?

(A) They can never be accessed by subclasses that extend the class.
(B) They are not visible to other objects.
(C) They cannot be used in a support class.
(D) Their values cannot be modified.
(E) They are shared by all instances of the class.

14. Which of the following is not true of try..catch blocks?

(A) They are a mechanism for handling exceptions (errors).
(B) Code which may cause an exception should be placed in a try block.
(C) Code which handles an exception should be placed in a catch block.
(D) A try block may have more than one catch block associated with it.
(E) A catch block may have more than one try block associated with it.

15. If class B extends class A, what members does an instance of class B have?

(A) Members declared in B.
(B) Members declared in B and all members declared in A.
(C) Members declared in B and private members declared in A.
(D) Members declared in B and static members declared in A.
(E) Members declared in B and non-private members declared in A.

16. Which of the following is not true of inner classes?

(A) All members of an outer class are visible to any inner class that it contains.
(B) An inner class can be used only within the outer class that contains it.
(C) The data fields of an outer class must be declared as public.
(D) An outer class can create instances of an inner class.
(E) Methods in the inner class can directly access members of the outer class.

17. Assume a variable called scan of type Scanner which is constructed so as to read from a file. What is the effect of the code below?

```
String s;
while ( scan.hasNext() ) {
   s = scan.nextLine();
   System.out.println( s );
}
```

(A) It prints out the tokens in the file one at a time on separate lines.
(B) It prints out the tokens on the first line of the file.
(C) It prints out the tokens on the last line of the file.
(D) It prints out each line of the file in order.
(E) It prints out every second line of the file.

18. A shadowing relationship between data fields is like what kind of relationship between methods?

    (A) overloading
    (B) overriding
    (C) returning
    (D) calling
    (E) weak polymorphism

19. Which of the following statements is true of both an abstract class and an interface?

    (A) They can both contain abstract methods.
    (B) They can both be extended by subclasses.
    (C) They can only contain public static final data fields.
    (D) They can both contain ordinary methods.
    (E) They can only be implemented by listener classes.

20. Which of the following statements about packages is **false**?

    (A) A class that has not been imported can be referred to using its fully qualified name.
    (B) Classes that are not explicitly put in a package are automatically put in the default package
    (C) By default the members of a class are visible to other classes in the same package.
    (D) The package java.lang is automatically imported into every Java program.
    (E) The header of a class contains information about its package.

21. Which of the statements regarding the Java event model is not correct?

    (A) Events are generated by component objects.
    (B) Each type of event is represented by an event class.
    (C) For every event class there is a corresponding listener interface.
    (D) For every listener interface there is a corresponding container object.
    (E) When an event is fired an event object is created and passed to objects that are listeners for that event.

22. Which of the following components is best suited to allowing the user to indicate a “true vs. false” selection?

(A) JPanel
(B) JButton
(C) JCheckbox
(D) JLabel
(E) JTextArea

23. Which of the following is the typical way of approaching the design of a program that is to be controlled by a GUI?

(A) As a set of functions applied to input data.
(B) As a specification of how to respond to external events.
(C) As a logical formula whose correctness must be proved.
(D) As a complete specification of a structured algorithm.
(E) As a user centered virtual reality environment.

24 What are "heavyweight" graphics classes?

(A) Classes which are involved in 3D graphics and modelling.
(B) Classes which perform very complex operations that display slowly.
(C) Classes which the programmer creates by extending existing graphics classes.
(D) Classes which are specifically optimised for particular graphics cards.
(E) Classes which interact directly with the underlying operating system.

Short answer section.

Please fill in your ID number on the top of all short answer pages, and write answers in the spaces provided in this script.

25. Describe two of the supposed benefits of an object–oriented approach to programming. (1)

(a) Robust/Safe

(b) Reuse, avoid repetition

26. Briefly describe each of the following kinds of error, and note which tool in the JDK detects the error and generates the error message for the programmer. (2)

syntax error: When the syntax in the code is wrong, the compiler detects it

runtime error: When the code breaks while the program is running

27. What are the two main packages of classes (or the general names for these packages) that are involved with creating graphical user interfaces (GUIs) in Java? (1)

javax.swing.*; java.awt.*;

29. Given the variables shown, what does each of the following expressions evaluate to? (4)

```
int  x = 1, y = 2;
double  d = 2.5;
String s = "The end";
```

d > 3.0 && x < y	false
d > 3.0 || x < y	true
!(d > 3.0) && x < y	true
!(d > 3.0 || x < y)	false
s.charAt(2)	e
s + s	The endThe end
(s + s).length()	14
s.substring(0, s.indexOf('e'))	Th

30. Write statements to: (1)

declare (but not initialise) a variable of type array of int, int[] x;

declare and initialise an array of double that can hold 50 values. double[] y = new double[50];

31. Write a method called maximum which accepts three integer values as input parameters. The body of the method should use an if..else..if sequence to return the maximum value or 0 if no single maximum can be determined (e.g. in the case of inputs such as 3, 7, 7). (3)

```
public int maximum(int x, int y, int z){
    if (x > y && x > z){
        return x;
    }
    else if(y > x && y > z){
        return y;
    }
    else if(z > x && z > y){
        return z;
    }
    else{
        return 0;
    }
}
```

32. Given a variable arr referring to a fully initialised array of int, write a while loop that prints out the values in the array. Include any declarations necessary. (2)

```
int count = 0;
while (count < arr.length){
    System.out.println(arr[count]);
    count++;
}
```

33. To the initial code on the next page, add the following:

(a) A constructor for the class Parent which takes as inputs two integers and uses them to set the values of the data fields x and y. (2)

(b) A replacement for the default constructor for the class Parent. (1)

(c) A new class Child which is a subclass of Parent, and has a single data field d of type double. (2)

(d) A constructor for the class Child which takes inputs for all the data fields of a Child instance object. Where appropriate the constructor in the superclass should be called. (2)

(e) A toString method for the class Child which returns a description of the values stored in the data fields of a Child instance object. (2)

```
public class Parent {

    protected int x, y;

    public Parent(){}

    public Parent(int x, int y){
        this.x = x;
        this.y = y;
    }



} // end of class Parent
```

```
public class Child extends Parent{
    private double d;

    public Child(int x, int y, double d){
        super(x, y);
        d = d;
    }
    public String toString(){
        return ("x = " + super.x + " y = " + super.y + " d = " + d)
    }
}
```

34. Given the declaration of the array below, write nested for loops that create the output shown to the right. Note that the array has rows of unequal length. (2)

```
int [ ] [ ] data = { {4, 7, 2}, {3, 9}, {8, 8, 6} };
```

Output

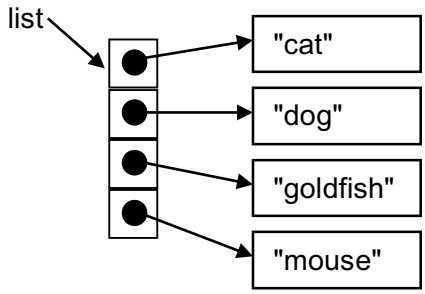
```
4 7 2
3 9
8 8 6
```

```
for (int x = 0; x < data.length; x++){
    for (int y = 0; y < data[x].length; y++){
      System.out.print(data[x][y] + " ");
    }
    System.out.println();
  }
```

35. Given the same array as Question 34 above, write nested for-each loops that create the same output. (2)

```
for (int[] x: data){
   for (int y: x){
     System.out.print(y + " ");
   }
   System.out.println();
 }
```

36. Assume an ArrayList called list with the initial structure shown in the model below. Sketch the new model of list after the code (which follows the model) has run. (2)




```
list.remove(1);
list.add("python");
list.add(2, "rabbit");
```

```
cat
goldfish
rabbit
mouse
python
```

37. Assume a main method that creates an instance of class B, and calls the test method on it. In the spaces to the right below, show the six lines of output that are printed. (3)

```
public class A {
  protected int a = 2;

  public void fruit( ) {
    System.out.println("Apple");
  }
}
```

```
public class B extends A {
  private int a = 3;

  public void fruit() {
    System.out.println("Banana");
  }

  public void fruit(int x) {
    System.out.println("Banana " + x);
  }

  public void fruit(int x, int y) {
    System.out.println(y + " Banana " + x);
  }

  public void test( ) {
   int a = 4;
   System.out.println(a);
   System.out.println(this.a);
   System.out.println(super.a);
   fruit(7);
   this.fruit();
   super.fruit();
  }
}
```

4

3

2

Banana 7

Banana

Apple

38. The code below reads a value, which should be an int, from the console (i.e. which is typed in by the user). If the user enters any other kind of value then an InputMismatchException is generated and the program crashes. Add the further code that is required to ensure that the program handles this error and instead of crashing prints out a warning to the user. (2)

```
try{
    Scanner s = new Scanner(System.in);
    System.out.println("Please enter an int");
    input = s.nextInt();

}
catch(InputMismatchException e){
   System.out.println("That wasn't an int");
}
```

39. Briefly answer the following: (2)

What class is the root of the Java class hierarchy?

Object

Any constructor either explicitly or automatically calls the constructor of its parent class, which calls its parent, and so on up the class hierarchy. What is the name of this process?

Inheritance

What is the point of the process described above?

To avoid having to write the same code over and over

Some languages allow a class to have more than one parent. What is the term that is usually used to describe such a design?

Multiple Inheritance

40. Assume an application class as shown, and a worker class AClass with an int data field set by a constructor. When the program runs it creates four variables and constructs three objects, as shown in the partial model to the right. When the program reaches the point of printing the message "Hello", the variables w, x, y, and z refer to various objects.

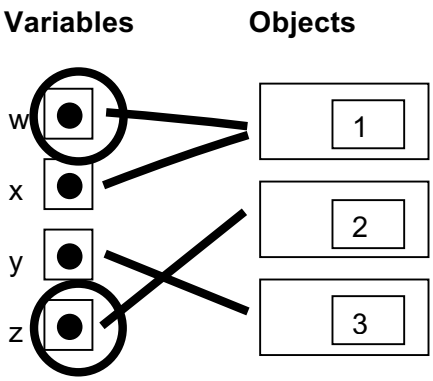
(a) Show the correct references at this point by adding arrows from the variables to objects. (2)

(b) In the model, circle the variables that are accessible within the method sayHello. (1)

```
public class Demo {
  private static AClass w = new AClass(1);

  public static void main(String [ ] args) {
      AClass x = w;
      AClass y = new AClass(3);
      sayHello( new AClass(2) );
  }

  public static void sayHello ( AClass z ) {
      System.out.println("Hello");
  }
}
```



TURN OVER

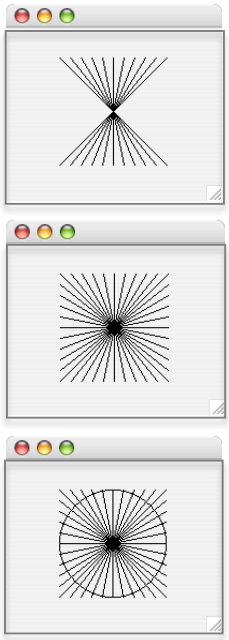
41. Creating an instance of the MyFrame class below results in graphical output displayed in a window, as shown in the first (topmost) of the three images below.

What single line of code could replace comment 1, so as to result in the output displayed in the second image? (2)

g.drawLine(left, top + i, right, bottom - i);

What single line of code could replace comment 2, so as to add the circle displayed in the third image? (1)

g.drawOval(top, left, right - left, bottom - top);



```
import java.awt.*;
import javax.swing.*;

public class MyFrame extends JFrame {

  public void paint (Graphics g) {

      int top = 50;
      int bottom = 150;
      int left = 50;
      int right = 150;
      for (int i = 0; i <= 100; i += 10) {
            g.drawLine(left + i, top, right - i, bottom);
            // comment 1
      }
      // comment 2
  }

}
```

42. Given the following classes: (1)

Vehicle

Car **extends** Vehicle

Toyota **extends** Car

Starlet **extends** Toyota

What are **all** the legal types of an object which is an instance of Toyota?

Toyota, Car, Vechile

43. Assume an application class which displays a JFrame (window). The only contents of (the content pane of) the JFrame is an instance of SliderPanel below, hence the contents of SliderPanel specify the behaviour of the program. The program displays the current numerical value of a slider in the range 0 to 99 (e.g. the image shows the GUI after the slider has been set to the position with the value 20). The following 5 questions refer to the lines in the program identified by comments (question 1 to comment 1, 2 to comment 2, and so on).

(1) What two modifiers will have been used in the declaration of the data field red ? (1)

public, static

(2) What aspect of event handling is implemented by calling addChangeListener ? (1)

Registering a listener

(3) What kind of value is passed as an input to the setPreferredSize method? (1)

Reference to an object of type Dimension

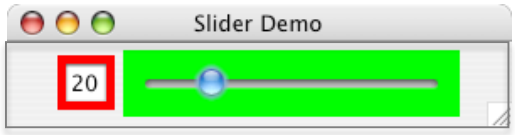
(4) How is it that SliderPanel has an add method? (1)

Inheritence

(5) What is ChangeListener, and what constraint does implementing it impose? (1)

Interface. Must specify a method called stateChanged

In the box at the lower right, draw the containment hierarchy for the complete program (i.e. including the JFrame created by the application class). (2)



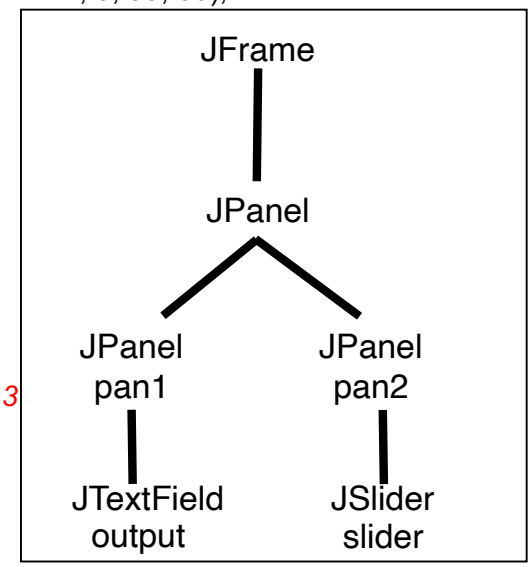

```
import java.awt.*;  import javax.swing.*;
import java.awt.event.*;  import javax.swing.event.*;

public class SliderPanel extends JPanel {
  private JTextField output = new JTextField("50");
  private JSlider slider = new JSlider(JSlider.HORIZONTAL, 0, 99, 50);
  private JPanel pan1 = new JPanel();
  private JPanel pan2 = new JPanel();

  public SliderPanel () {
    pan1.setBackground(Color.red);  // comment 1
    pan1.add(output);
    pan2.setBackground(Color.green);
    MyListener ml = new MyListener();
    slider.addChangeListener(ml);  // comment 2
    pan2.add(slider);
    setPreferredSize(new Dimension(300, 50)); // com 3
    add(pan1);  // comment 4
    add(pan2);
  }

  private class MyListener implements ChangeListener {   // comment 5
    public void stateChanged(ChangeEvent e) {
      output.setText("  " + slider.getValue() );
    }
  } // MyListener

} // SliderPanel
```

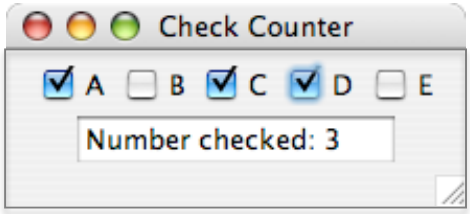


TURN OVER

44. Assume an application class which displays a JFrame (window). The only contents of (the content pane of) the JFrame is an instance of CounterPanel below, hence the contents of CounterPanel specify the behaviour of the program. For example, the image below shows the JFrame after check boxes A, C and D have been checked.

Complete the code below so that the program behaves as shown in the image, where the text in the JTextField describes the number of check boxes checked (ticked). (5)

Note: JCheckBoxes have an isSelected method that returns true if the check box is checked.




```
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;

public class CounterPanel extends JPanel {

  private JCheckBox [ ] boxes = {
    new JCheckBox("A"), new JCheckBox("B"),
    new JCheckBox("C"), new JCheckBox("D"),
    new JCheckBox("E")
  };

  private JTextField output = new JTextField(12);

  public CounterPanel() {

    MyListener ml = new MyListener();

   for (JCheckBox b : boxes) {  // for each box register listener and add to panel

      b.addActionListener(ml);
      add(b);

    }

    setPreferredSize( new Dimension(220, 75) );

    add(output);

  }

  private class MyListener implements ItemListener {

    public void itemStateChanged (ItemEvent event) {

      int count = 0;

        for(JCheckbox b: boxes){
           if (b.isSelected()) count ++;
        }

      output.setText("Number checked: " + count);

    }

  }

} // end of Counter
```